Recall : Systems of Linear Equations

- A system of linear equations is a collection of equations in variables $x_1, ..., x_n$ of the form

$$a_1x_1 + \cdots + a_nx_n = b.$$

Example.

$$\begin{cases} x_1 & +x_2 & -x_3 & = 1 \\ x_1 & & +x_3 & = -1 \\ & -x_2 & +2x_3 & = 0 \end{cases}$$

- We are interested in finding the solution set of a system of linear equations.
- We saw that Gaussian elimination is realized by elementary row operations on the augmented matrix.

Recall : augmented matrix

$$\begin{cases} x_1 & +x_2 & -x_3 & = 1 \\ x_1 & & +x_3 & = -1 \\ & -x_2 & +2x_3 & = 0 \end{cases} \rightsquigarrow \left[\begin{array}{ccc|c} 1 & 1 & -1 & 1 \\ 1 & 0 & 1 & -1 \\ 0 & -1 & 2 & 0 \end{array}\right]$$

$x_1 \quad x_2 \quad x_3$

augmented matrix

• To solve a linear system, we use elementary row operations to put the augmented matrix into the reduced echelon form (R.E.F.).

$R_i \to R_i + cR_j$ replacement

$R_i \leftrightsquigarrow R_j$ interchange

$R_i \to cR_i,\ c \neq 0$ scaling

Example.

$$\begin{bmatrix} 1 & -1 & 0 & 2 \\ 0 & 1 & -2 & 0 \\ 2 & 0 & -1 & 1 \end{bmatrix} \to \begin{bmatrix} 1 & -1 & 0 & 2 \\ 0+2 & 1+(-2) & -2+0 & 0+4 \\ 2 & 0 & -1 & 1 \end{bmatrix} \quad R_2 \to R_2 + 2R_1$$

$$\begin{bmatrix} 1 & -1 & 0 & 2 \\ 0 & 1 & -2 & 0 \\ 2 & 0 & -1 & 1 \end{bmatrix} \to \begin{bmatrix} 1 & -1 & 0 & 2 \\ 2 & 0 & -1 & 1 \\ 0 & 1 & -2 & 0 \end{bmatrix} \quad R_2 \leftrightsquigarrow R_3$$

$$\begin{bmatrix} 1 & -1 & 0 & 2 \\ 0 & 1 & -2 & 0 \\ 2 & 0 & -1 & 1 \end{bmatrix} \to \begin{bmatrix} 1 & -1 & 0 & 2 \\ 0 & -2 & 4 & 0 \\ 2 & 0 & -1 & 1 \end{bmatrix} \quad R_2 \to -2R_2$$

Two matrices are called row equivalent if one can get from one to the other by a sequence of elementary row operations.

Important Fact

Given two linear systems, if their augmented matrices are row equivalent, then their solution set is the same.

This means that if we can solve one of them, then we have solved the other as well.

- To solve a linear system, we first use elementary row operations to obtain the **reduced echelon form** (R.E.F.) of its augmented matrix.

Echelon form (E.F.)

The leading entry of a row of any matrix is the first nonzero entry from the left in that row.

Example:

$$\begin{bmatrix} 0 & \mathbf{-3} & 2 & 4 \\ 0 & 0 & \mathbf{2} & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

In the above example, the boldface numbers are the leading entries of the first two rows. The third row does not have a leading entry.

- We say that a matrix is in echelon form if it satisfies the following properties:

1. All nonzero rows are above all zero rows.
2. Each leading entry lies to the right of any leading entry above it.
3. All entries in a column below a leading entry are zeros.

Reduced Echelon Form

We say a matrix is in reduced echelon form (R.E.F.) if it is in E.F. and satisfies the following conditions:

1. All of its leading entries are equal to 1 (and therefore we will call them leading 1's).
2. Each leading 1 is the only nonzero entry of the column that contains it.

Example. $\begin{bmatrix} 0 & 1 & -1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & -4 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$ is in R.E.F.

Pivot positions, pivot columns

When a matrix is in E.F. (and in particular, in R.E.F.):

- The location of a leading 1 is called a pivot position.
- A column containing a pivot position is called a pivot column.

The row reduction algorithm

Find the E.F. of $\begin{bmatrix} 0 & 0 & 1 \\ -1 & 0 & 3 \\ 1 & 0 & -3 \\ -1 & 0 & 1 \end{bmatrix}$

1. Start with the leftmost nonzero column. This is a pivot column. The pivot position of this column will be the topmost entry.
2. Select a nonzero entry in this column. If necessary, use an interchange $R_1 \leftrightsquigarrow R_k$ to move this nonzero entry to the pivot position.

$R_1 \leftrightsquigarrow R_2$ $\begin{bmatrix} -1 & 0 & 3 \\ 0 & 0 & 1 \\ 1 & 0 & -3 \\ -1 & 0 & 1 \end{bmatrix}$

The row reduction algorithm (cont.)

Last matrix from previous slide : $\begin{bmatrix} -1 & 0 & 3 \\ 0 & 0 & 1 \\ 1 & 0 & -3 \\ -1 & 0 & 1 \end{bmatrix}$

3. Using replacements ($R_i \to R_i + cR_j$) create zeros in all positions below the pivot position.

$$\begin{bmatrix} -1 & 0 & 3 \\ 0 & 0 & 1 \\ 1 & 0 & -3 \\ -1 & 0 & 1 \end{bmatrix} \xrightarrow{R_3 \to R_3 + R_1} \begin{bmatrix} -1 & 0 & 3 \\ 0 & 0 & 1 \\ \mathbf{0} & \mathbf{0} & \mathbf{0} \\ -1 & 0 & 1 \end{bmatrix} \xrightarrow{R_4 \to R_4 - R_1} \begin{bmatrix} -1 & 0 & 3 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \\ \mathbf{0} & \mathbf{0} & \mathbf{-2} \end{bmatrix}$$

• We are done with the first pivot. We now find the second, third,...pivot.

The row reduction algorithm (cont.)

Last matrix on previous slide: $\begin{bmatrix} -1 & 0 & 3 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \\ \mathbf{0} & \mathbf{0} & \mathbf{-2} \end{bmatrix}$

4. Ignore the row containing the current pivot position and all rows that lie above it.

$$\begin{bmatrix} \mathbf{-1} & 0 & 3 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & -2 \end{bmatrix}$$

Repeat the process for the smaller matrix.
(If there are no nonzero entries left, then stop! You have obtained an E.F.)

$$\begin{bmatrix} \mathbf{-1} & 0 & 3 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & -2 \end{bmatrix} \xrightarrow{R_4 \to R_4 + 2R_2} \begin{bmatrix} -1 & 0 & 3 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \text{E.F.}$$

The row reduction algorithm (cont.)

6. (to obtain R.E.F. from E.F.)
Beginning with the rightmost pivot and working upward and to the left, create zeros above each pivot (using replacement operations).
- If a pivot is not 1, make it 1 by a scaling operation.

Last matrix on the previous slide : $\begin{bmatrix} -1 & 0 & 3 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

$$\begin{bmatrix} -1 & 0 & 3 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_1 \to R_1 - 3R_2} \begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_1 \to -R_1} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$ R.E.F.

Summary of the algorithms

The following algorithm computes an E.F. of any given matrix.

Algorithm to compute the E.F.

- Find the first nonzero column from the left. If there are no nonzero columns left, then stop.
- The top position of this column is a *pivot position*.
- If this pivot position contains a zero, use an interchange to make it nonzero.
- Use replacements (from top to bottom) to make all the entries below this pivot position (in the same column) equal to zero.
- Ignore the row containing the pivot position, and any other rows that possibly lie above that.
- Repeat the above steps on the remaining submatrix.

Summary of the algorithms

The following algorithm computes the R.E.F. of a matrix which is already in E.F.

Algorithm to compute R.E.F. from E.F.

- Start from the rightmost pivot position.
- If the value of the pivot position is not equal to 1, use a scaling of its row to make it equal to 1. Use replacements to make all entries above the pivot position (in the same column) equal to zero.
- Repeat the previous step for the pivot position that was found immediately before the current pivot position.

General solution of a linear system

Example

Solve the linear system $\begin{cases} -x_1 + 2x_2 + x_3 = 1 \\ 2x_1 - 4x_2 = -4 \\ -x_1 + 2x_2 + 3x_3 = -1 \end{cases}$

The augmented matrix is : $\left[\begin{array}{ccc|c} -1 & 2 & 1 & 1 \\ 2 & -4 & 0 & -4 \\ -1 & 2 & 3 & -1 \end{array}\right]$

- First we find the R.E.F. of the augmented matrix.

$$\left[\begin{array}{ccc|c} -1 & 2 & 1 & 1 \\ 2 & -4 & 0 & -4 \\ -1 & 2 & 3 & -1 \end{array}\right] \xrightarrow{R_2 \to R_2 + 2R_1} \left[\begin{array}{ccc|c} -1 & 2 & 1 & 1 \\ 0 & 0 & 2 & -2 \\ -1 & 2 & 3 & -1 \end{array}\right]$$

$$\xrightarrow{R_3 \to R_3 - R_1} \left[\begin{array}{ccc|c} -1 & 2 & 1 & 1 \\ 0 & 0 & 2 & -2 \\ 0 & 0 & 2 & -2 \end{array}\right] \xrightarrow{R_3 \to R_3 - R_2} \left[\begin{array}{ccc|c} \mathbf{-1} & 2 & 1 & 1 \\ 0 & 0 & \mathbf{2} & -2 \\ 0 & 0 & 0 & 0 \end{array}\right] \quad \text{E.F.} \quad \xrightarrow{R_2 \to \frac{1}{2}R_2}$$

$$\left[\begin{array}{ccc|c} -1 & 2 & 1 & 1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{array}\right] \xrightarrow{R_1 \to R_1 - R_2} \left[\begin{array}{ccc|c} -1 & 2 & 0 & 2 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{array}\right] \xrightarrow{R_1 \to -R_1} \left[\begin{array}{ccc|c} 1 & -2 & 0 & -2 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

R.E.F.

General solution of a linear system (cont.)

Example

Solve the linear system $\begin{cases} -x_1 & +2x_2 & +x_3 & = 1 \\ 2x_1 & -4x_2 & & = -4 \\ -x_1 & +2x_2 & +3x_3 & = -1 \end{cases}$

- Next we write the linear system whose augmented matrix is the obtained R.E.F.

$$\left[\begin{array}{ccc|c} 1 & -2 & 0 & -2 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{array}\right] \rightsquigarrow \begin{cases} x_1 & -2x_2 & & = -2 \\ & & x_3 & = -1 \\ & & 0 & = 0 \end{cases} \rightsquigarrow \begin{cases} x_1 = 2x_2 - 2 \\ x_2 : \text{ free} \\ x_3 = -1 \end{cases}$$

x_1 x_2 x_3

A basic variable is one corresponding to a pivot column.

A free variable is one which does not correspond to a pivot column.

- x_1 and x_3 are **basic** variables.
- x_2 is a **free** variable.

By solving the equations backwards, the basic variables are determined in terms of free variables.

General solution of a linear system (cont.)

Example

Solve the linear system $\begin{cases} -x_1 & +2x_2 & +x_3 & = 1 \\ 2x_1 & -4x_2 & & = -4 \\ -x_1 & +2x_2 & +3x_3 & = -1 \end{cases}$

To check your solution, you plug in the values into equations and make sure all of them are satisfied.

$$\begin{cases} x_1 = 2x_2 - 2 \\ x_2 = \text{free} \\ x_3 = -1 \end{cases} \rightsquigarrow \begin{cases} -(2x_2 - 2) & +2x_2 & -1 & = 1 \\ 2(2x_2 - 2) & -4x_2 & & = -4 \\ -(2x_2 - 2) & +2x_2 & +3(-1) & = -1 \end{cases}$$

- $-x_1 + 2x_2 + x_3 = -(2x_2 - 2) + 2x_2 + (-1) = -2x_2 + 2 + 2x_2 - 1 = 1.$ ✓
- $2x_1 - 4x_2 = 2(2x_2 - 2) - 4x_2 = 4x_2 - 4 - 4x_2 = -4.$ ✓
- $-x_1 + 2x_2 + 3x_3 = -(2x_2 - 2) + 2x_2 + 3(-1) = -2x_2 + 2 + 2x_2 - 3 = -1$✓

Examples of solutions:

$x_1 = 0, x_2 = 1, x_3 = -1$

$x_1 = -4, x_2 = -1, x_3 = -1$

$x_1 = -1, x_2 = \frac{1}{2}, x_3 = -1$

To solve a linear system...

1. Find the R.E.F. of the augmented matrix of the system.
2. Write the linear system whose augmented matrix is the R.E.F.
3. You can easily solve the latter system backwards. The basic variables can be expressed in terms of the free variables.

Important Remarks

- A matrix can have several different E.F.'s.

  Example. $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ and $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ are row equivalent.

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \xrightarrow{R_1 \to R_1 - R_2} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

- The R.E.F. of a matrix is unique.
- It is also possible to use the E.F. to solve a linear system, BUT ...
- You are strongly advised to use the R.E.F., because there is less chance of numerical error.

Another Example

Example

Solve the linear system $\begin{cases} x_1 + 2x_2 - x_4 = 1 \\ x_1 + 2x_2 - x_3 + x_4 = -2 \\ -x_1 - 2x_2 + 2x_3 - 3x_4 = 5 \end{cases}$

Augmented matrix: $\left[\begin{array}{cccc|c} 1 & 2 & 0 & -1 & 1 \\ 1 & 2 & -1 & 1 & -2 \\ -1 & -2 & 2 & -3 & 5 \end{array}\right]$

$$\left[\begin{array}{cccc|c} 1 & 2 & 0 & -1 & 1 \\ 1 & 2 & -1 & 1 & -2 \\ -1 & -2 & 2 & -3 & 5 \end{array}\right] \xrightarrow[R_3 \to R_3 + R_1]{R_2 \to R_2 - R_1} \left[\begin{array}{cccc|c} 1 & 2 & 0 & -1 & 1 \\ 0 & 0 & -1 & 2 & -3 \\ 0 & 0 & 2 & -4 & 6 \end{array}\right]$$

$$\xrightarrow{R_3 \to R_3 + 2R_2} \left[\begin{array}{cccc|c} 1 & 2 & 0 & -1 & 1 \\ 0 & 0 & -1 & 2 & -3 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right] \text{ E.F. } \xrightarrow{R_2 \to -R_2} \left[\begin{array}{cccc|c} 1 & 2 & 0 & -1 & 1 \\ 0 & 0 & 1 & -2 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right] \text{ R.E.F.}$$

The last matrix is the augmented matrix of $\begin{cases} x_1 + 2x_2 - x_4 = 1 \\ x_3 - 2x_4 = 3 \\ 0 = 0 \end{cases}$

Another Example (cont.)

Example

Solve the linear system $\begin{cases} x_1 & +2x_2 & & -x_4 & = 1 \\ x_1 & +2x_2 & -x_3 & +x_4 & = -2 \\ -x_1 & -2x_2 & +2x_3 & -3x_4 & = 5 \end{cases}$

$$\left[\begin{array}{cccc|c} 1 & 2 & 0 & -1 & 1 \\ 0 & 0 & 1 & -2 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

x_1 x_2 x_3 x_4

x_1, x_3 are basic and x_2, x_4 are free.

$$\begin{cases} x_1 & +2x_2 & & -x_4 & = 1 \\ & & x_3 & -2x_4 & = 3 \\ & & & 0 & 0 \end{cases} \quad \rightsquigarrow \quad \text{General solution:} \quad \begin{cases} x_1 = -2x_2 + x_4 + 1 \\ x_2 : \text{ free} \\ x_3 = 2x_4 + 3 \\ x_4 : \text{ free} \end{cases}$$

We will now check this solution.

Another Example (cont.)

Example

Solve the linear system $\begin{cases} x_1 & +2x_2 & & -x_4 & = 1 \\ x_1 & +2x_2 & -x_3 & +x_4 & = -2 \\ -x_1 & -2x_2 & +2x_3 & -3x_4 & = 5 \end{cases}$

General solution : $\begin{cases} x_1 = -2x_2 + x_4 + 1 \\ x_2 : \text{ free} \\ x_3 = 2x_4 + 3 \\ x_4 : \text{ free} \end{cases}$

- $x_1 + 2x_2 - x_4 = (-2x_2 + x_4 + 1) + 2x_2 - x_4 = 1$ $\surd$
- $x_1 + 2x_2 - x_3 + x_4 = (-2x_2 + x_4 + 1) + 2x_2 - (2x_4 + 3) + x_4 = -2$ $\surd$
- $-x_1 - 2x_2 + 2x_3 - 3x_4 = -(-2x_2 + x_4 + 1) - 2x_2 + 2(2x_4 + 3) - 3x_4 = 5$ $\surd$

Existence and Uniqueness

- When does a system have a unique solution?
- When does a linear system have infinitely many solutions?
- When does a linear system have no solutions?

First we calculate an E.F. of the augmented matrix of the system.

- If there is a row of the form

$$[\,0 \;\; 0 \;\; \cdots \;\; 0 \;\; 0 \mid b\,]$$

where $b \neq 0$ then the linear system is inconsistent (i.e. has no solutions).

- Equivalently, if the **rightmost column** of the augmented matrix is a pivot column, then the system is inconsistent.

Otherwise, the linear system is consistent (i.e., has solutions).

For a consistent linear system :

- If there is at least one **free** variable, then there are infinitely many solutions.
- If all the variables are **basic**, then the system has a unique solution.

Summary of today's lecture

- To find the general solution of a linear system:
  1. Find the R.E.F. of the augmented matrix. Determine if the system is consistent or inconsistent.
  2. If the system is consistent, write a new linear system whose augmented matrix is the computed R.E.F. In each equation, there is a **unique** basic variable. By isolating that variable, find a formula for it in terms of free variables.
- In the general solution of a consistent system, the basic variables are expressed in terms of the free variables.
- The procedure for solving a linear system is routine. It does not involve any tricks.

Yes, you can do it!